

FITBUDDY - AI FITNESS PLAN GENERATOR USING GEMINI MODELS

Phase 8: Project Demonstration Phase

The final phase focuses on presenting and demonstrating the completed FitBuddy application.

Demonstration Flow

The project demonstration can follow this sequence:

Step 1 – Home Page

Demonstrate the FitBuddy homepage and explain the user input fields.

Step 2 – Enter User Information

Enter:

- Name.
- User ID.
- Age.
- Weight.
- Fitness goal.
- Workout intensity.

Step 3 – Generate Workout Plan

Click **Generate Plan** and demonstrate how the application communicates with the Gemini-powered backend.

Step 4 – Display Personalized Plan

Show the generated 7-day workout plan, including exercises, sets, repetitions, warm-ups, and recovery guidance.

Step 5 – Display Nutrition Tip

Demonstrate the personalized nutrition/recovery recommendation.

Step 6 – Submit Feedback

Enter feedback such as requesting additional cardio, flexibility exercises, or other changes.

Step 7 – Generate Updated Plan

Demonstrate how the application uses the original plan and user feedback to generate an updated workout plan.

Step 8 – Admin View

Open the `/view-all-users` page and demonstrate how user information and original/updated plans are displayed.

Step 9 – API Documentation

Demonstrate FastAPI's `/docs` interface and available endpoints.

Step 10 – Explain Architecture

Conclude the demonstration by explaining the interaction between:

User → Jinja2 Frontend → FastAPI → Gemini AI → SQLite Database → Updated Response

Outcome:

A complete working demonstration showing the major features, AI integration, database functionality, frontend, backend, and feedback-based personalization of FitBuddy.